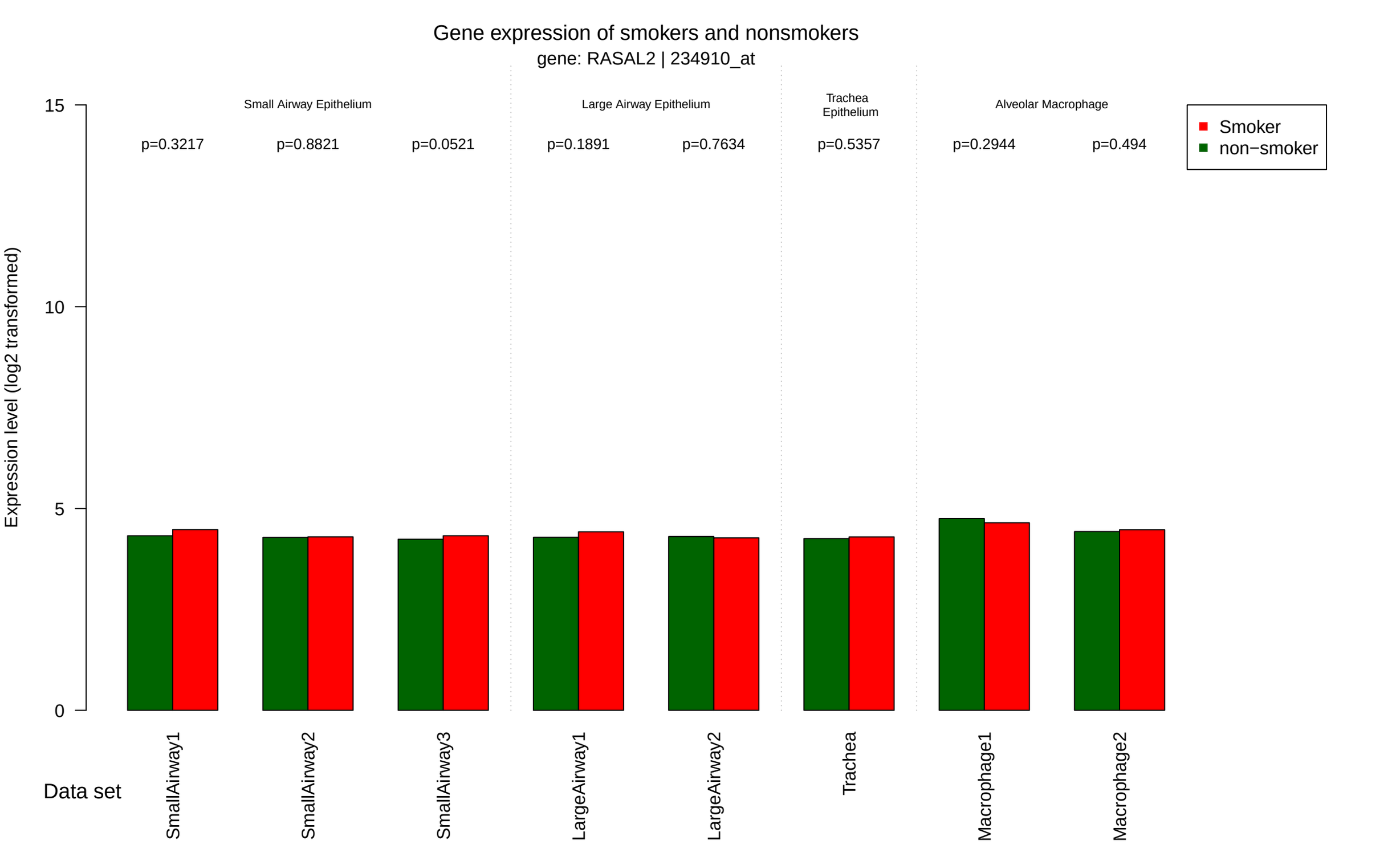
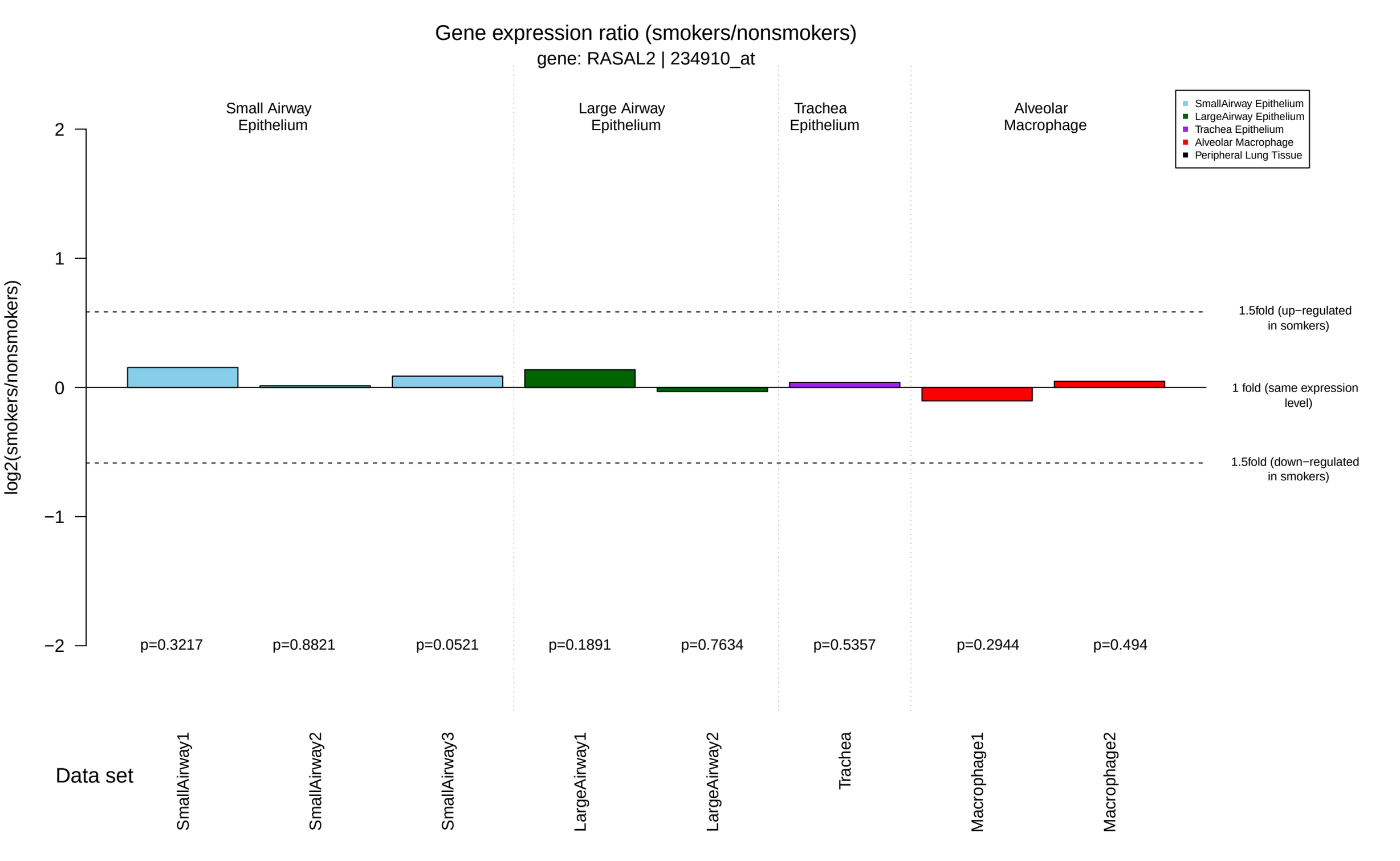
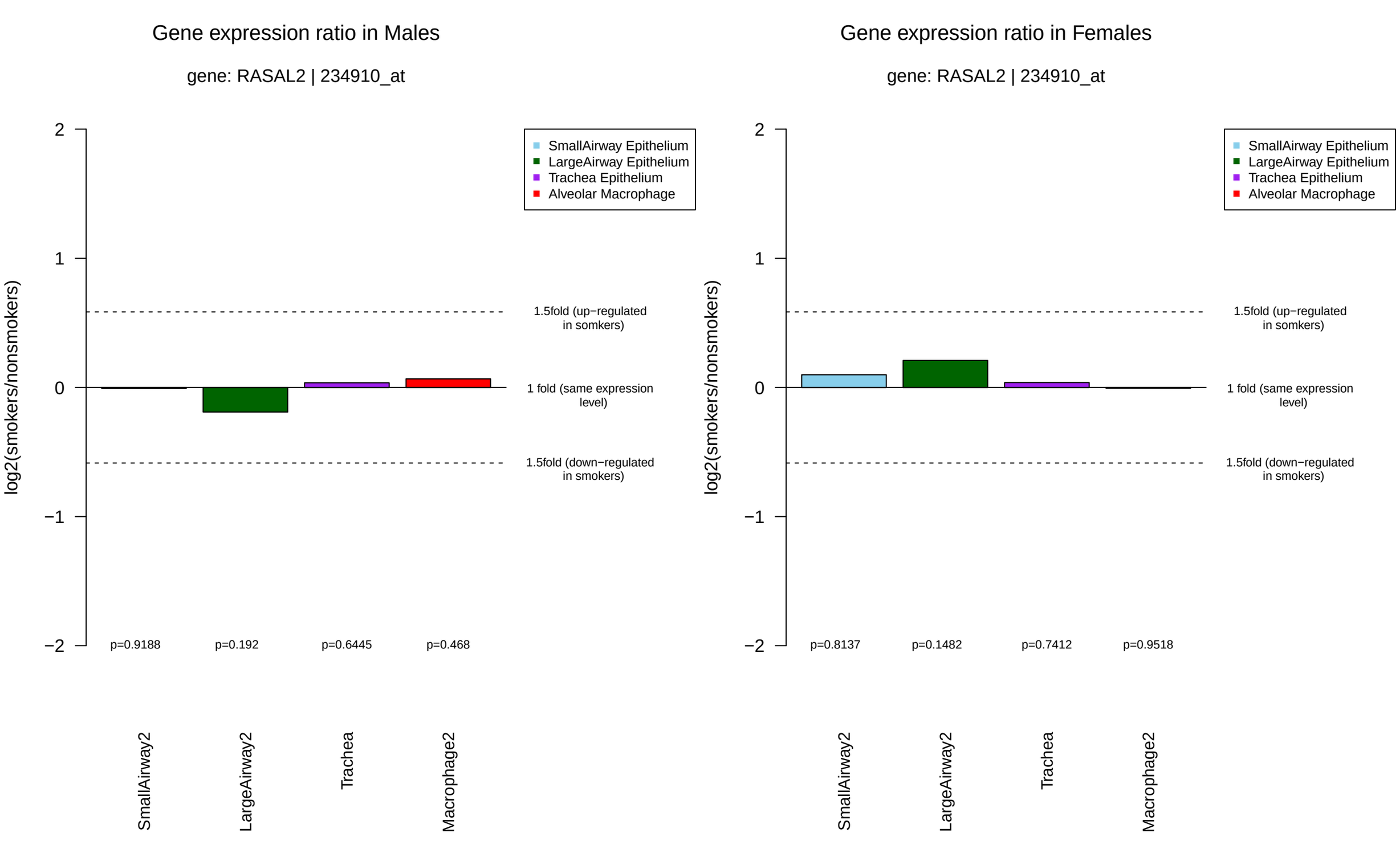
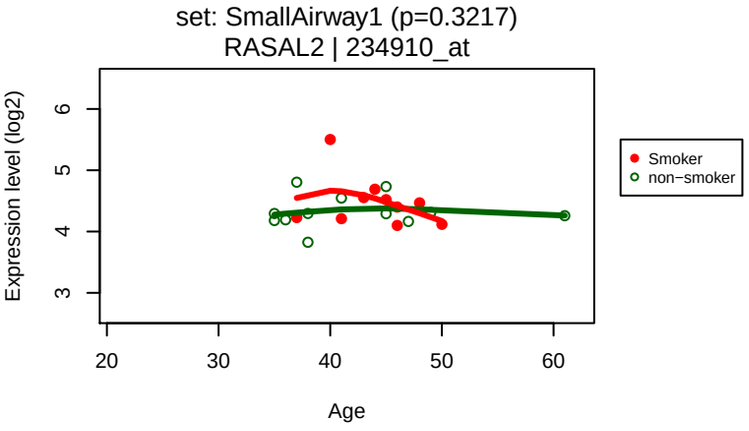
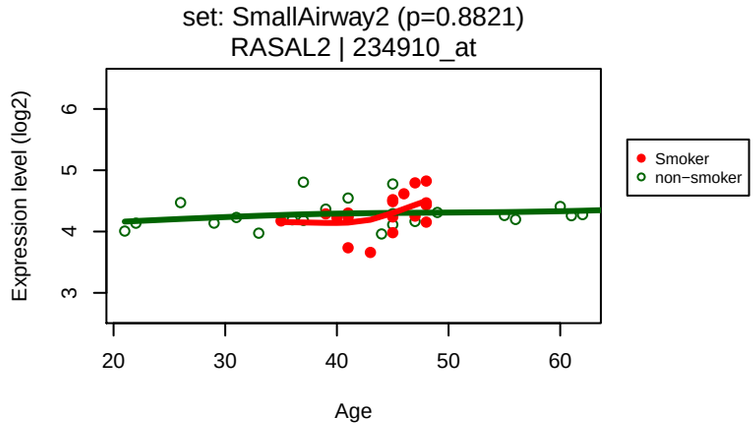
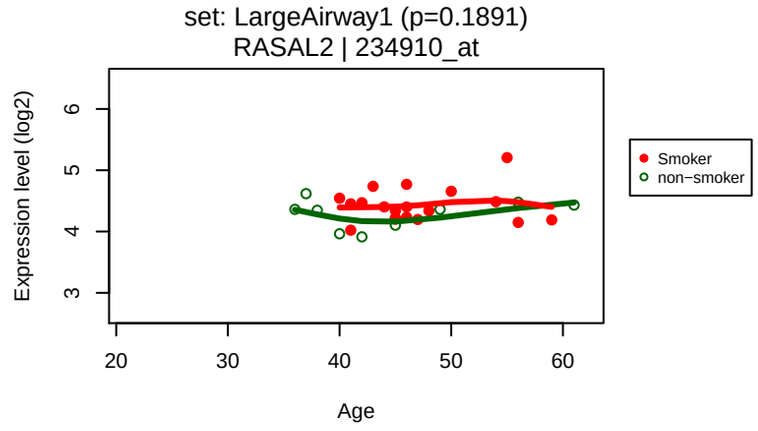

Gene expression of smokers and nonsmokers
gene: RASAL2 | 234910_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3217
p=0.8821
p=0.0521
p=0.1891
p=0.7634
p=0.5357
p=0.2944
p=0.494
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

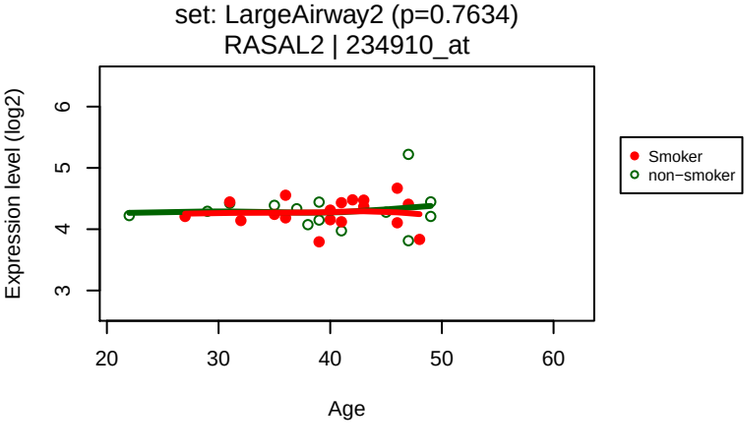

Gene expression ratio (smokers/nonsmokers)
gene: RASAL2 | 234910_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3217
p=0.8821
p=0.0521
p=0.1891
p=0.7634
p=0.5357
p=0.2944
p=0.494
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

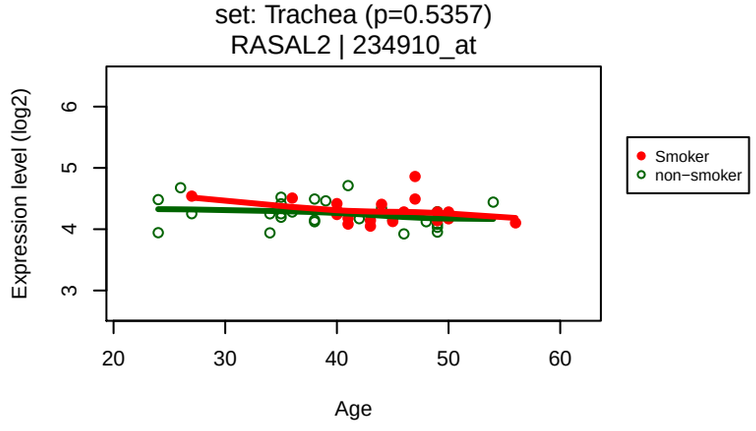

Gene expression ratio in Males
gene: RASAL2 | 234910_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9188
p=0.192
p=0.6445
p=0.468
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASAL2 | 234910_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8137
p=0.1482
p=0.7412
p=0.9518
SmallAirway2
LargeAirway2
Trachea
Macrophage2

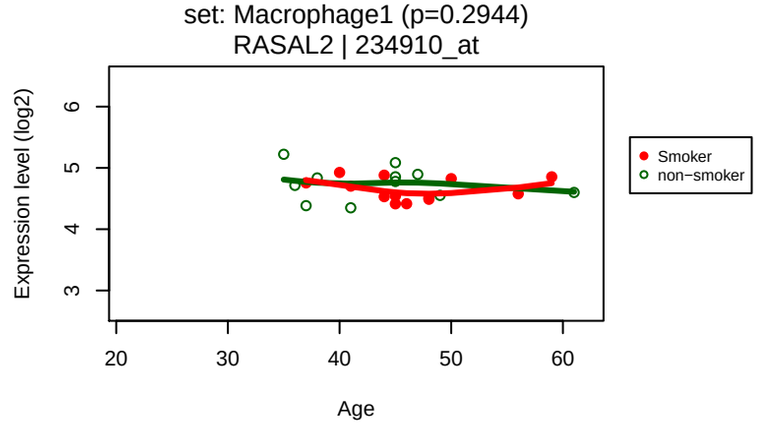

set: SmallAirway1 (p=0.3217)
RASAL2 | 234910_at
Expression level (log2)
Age
Smoker
non−smoker

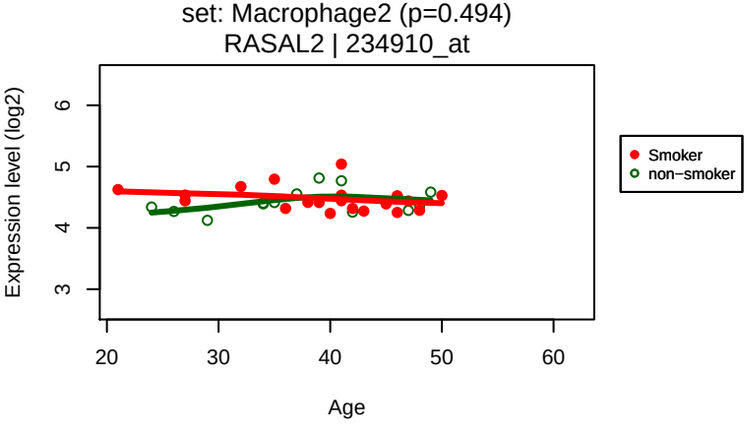

set: SmallAirway2 (p=0.8821)
RASAL2 | 234910_at
Expression level (log2)
Age
Smoker
non−smoker

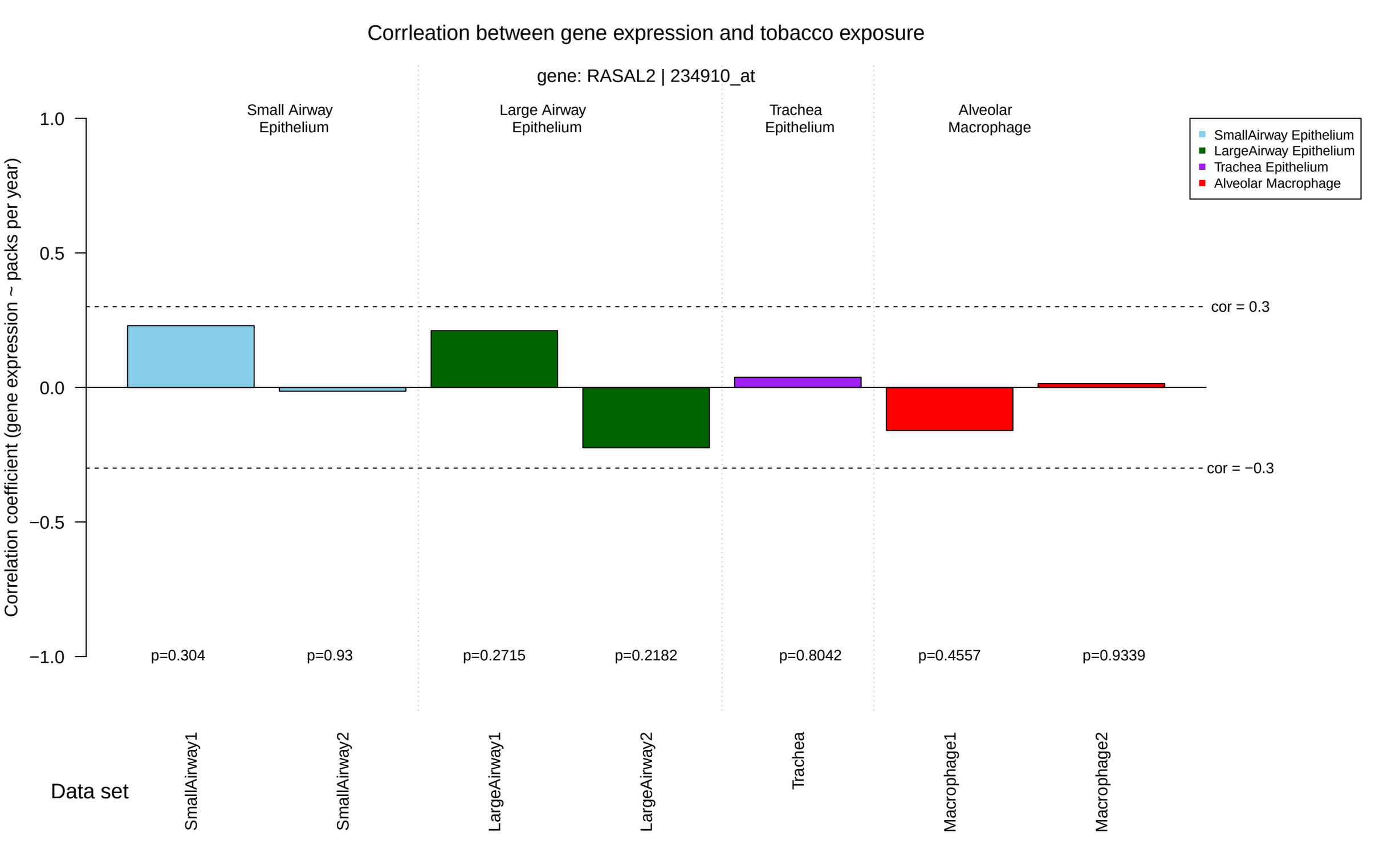

set: LargeAirway1 (p=0.1891)
RASAL2 | 234910_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7634)
RASAL2 | 234910_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5357)
RASAL2 | 234910_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2944)
RASAL2 | 234910_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.494)
RASAL2 | 234910_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASAL2 | 234910_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.304
p=0.93
p=0.2715
p=0.2182
p=0.8042
p=0.4557
p=0.9339
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2